

# MATH 2630-090: TEST 2 – Summer 2024

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Name (*please print*): \_\_\_\_\_

**Instructions:** Please read the following instructions carefully.

- **Read all of the directions** for each problem.
- You have **75 minutes** to finish the exam.
- **Show work to receive credit.** Answers that are wrong, but still have work might receive partial credit. If you provide the answer to a computation without work, I will not give partial credit.
- Provide **exact** solutions.
- Do **not** write in the table below.

| Problem      | Points | Score |
|--------------|--------|-------|
| 1            | 12     |       |
| 2            | 12     |       |
| 3            | 12     |       |
| 4            | 10     |       |
| 5            | 10     |       |
| 6            | 10     |       |
| 7            | 10     |       |
| 8            | 12     |       |
| 9            | 12     |       |
| <b>Total</b> | 100    |       |

**1. (12 points)** Find *all* first and second order partial derivatives of the function

$$f(x, y) = x^2 e^{xy}$$

**2. (12 points)** Find  $\frac{\partial z}{\partial t}$  and  $\frac{\partial z}{\partial s}$  at  $s = 1, t = 2, u = 3$ , if  $z = x^4 + x^2y$ ,  $x = s + 2t - u$ , and  $y = stu^2$ .

**3. (12 points)** Find *all* local maximum, minimum, and saddle point values of the function

$$f(x, y) = x^2 - xy + y^2 + 9x - 6y + 5.$$

4. **(10 point)** Find the derivative of  $f(x, y)$  in the direction of the vector  $u = \langle 2, -4 \rangle$ .

$$f(x, y) = 2x^3y - \frac{x}{y}$$

**5. (10 points)** Find the equation for the tangent plane and the normal line to the following surface at the point  $(1, 1, 1)$ .

$$yz + x \ln y = z^2.$$

**6. (10) points)** Evaluate the following limit. Write “*Does not exist*” if needed.

$$\lim_{(x,y) \rightarrow (0,0)} \frac{xy^3 + x^3y}{x^4 + y^4 + x^2y^2}.$$

**7. (10 points)** Consider the surface  $f(x, y) = \sin(2x + 2y)$ .

1. Find  $f_{xx}$
2. Find  $f_{xyy}$
3. Find  $f_{xyy \dots xyy}$  if there are  $k$ -copies of  $xyy$

**8. (12 points)** Consider the following 4-dimensional surface.

$$f(x, y, z) = 2x^3z + 3xyz + yz^2.$$

- (a) Find the gradient  $\nabla f(x, y, z)$ .
- (b) Find the direction of steepest ascent from the point  $f(1, 1, -1)$ .
- (c) Find the slope of the steepest ascent found in (b)

**9. (12 points)** Find the absolute maximum and minimum of the surface  $f(x, y) = x^2 + xy + y^2$  bounded by  $R = \{(x, y) : 0 \leq x \leq 1, 0 \leq y \leq 2\}$ .